SIGNIFICANCE

The unmet need. If we are fortunate, each of us will reach the point where we, or someone we love, needs help with the ordinary tasks of daily living: bathing, dressing, meals, medications, and moving safely through the home. Most families meet that moment at its worst, after a fall, a diagnosis, or a sudden hospitalization, and are expected to become experts in eldercare overnight.

A recognized need becomes established care only if a family clears three gates (Figure 1). They must **find** the right care inside a fragmented ecosystem of public aid programs, insurance coverage, healthcare services, and long-term services and supports, each with separate eligibility rules and none accountable for whether care begins. They must **afford** it: full-time home care runs $80,080 a year,[2] Medicare does not cover custodial home care, and Medicaid, the nation's largest payer of long-term care, pays only after a family has spent down to eligibility. An estimated $58 billion in assistance that would close part of that gap goes unclaimed every year because families cannot identify or navigate the programs.[1] And someone must be available to **deliver** the care: 63.3 percent of home-care providers declined cases in 2023 that they could not staff.[3]

Failure at any one gate produces the same outcome. Nearly a third of older adults with difficulty in activities of daily living go without bathing, meals, or medications in a given month,[4] and unmet needs of this kind independently predict emergency department use, readmission, nursing home placement, and mortality.[5,6,7]

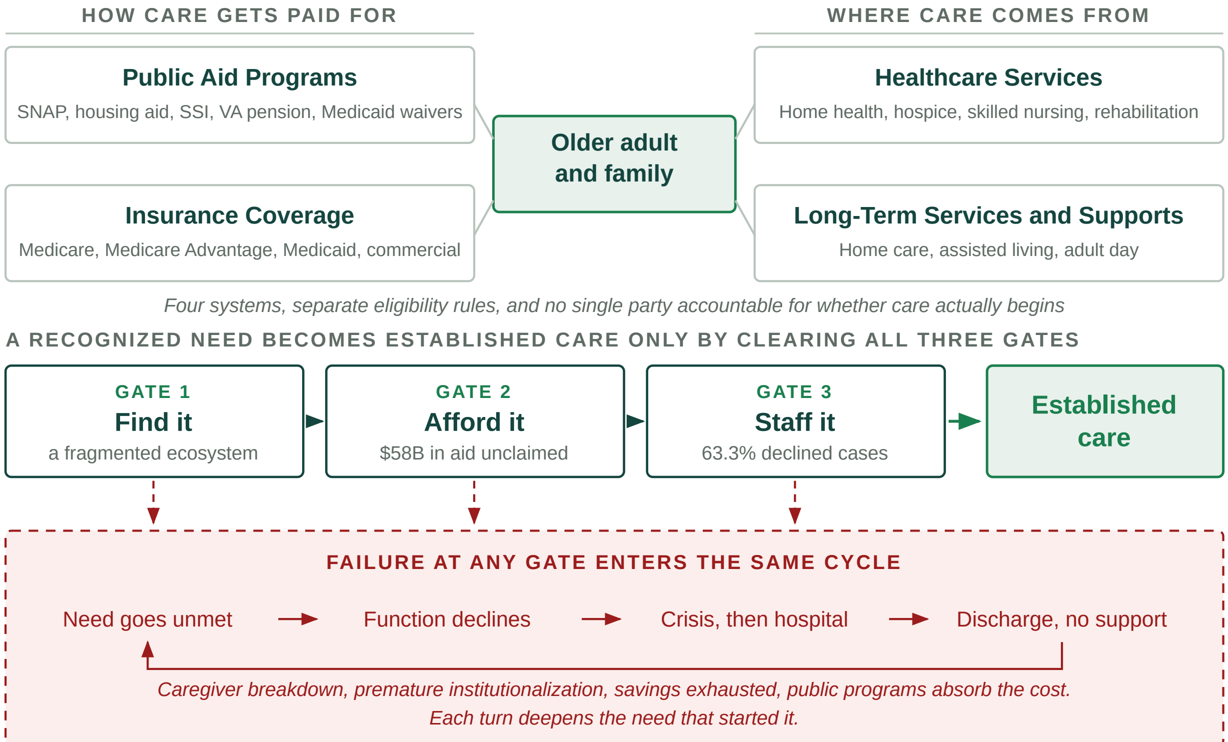

Figure 1. The eldercare ecosystem a family must navigate, the three gates that must all be cleared for a recognized need to become established care, and the cycle that follows failure at any one of them.

What it costs, and who pays it. The gap is widening from both sides. The population over 65 reaches 82 million by 2050,[8] while the family capacity that has always absorbed most of American eldercare is contracting: the number of potential family caregivers for each adult over 80 falls from more than seven in 2010 to four by 2030 (Figure 2).[9] The paid workforce that would have to replace it faces 9.7 million direct-care job openings between 2024 and 2034.[10] Demand is rising against two supplies that are falling at once.

Those costs do not disappear. They move, and they grow in transit. Roughly 70 percent of working-age family caregivers hold jobs, and the hours they lose are borne by employers and by their own earnings.[11] Families spend down their savings and Medicaid inherits the bill at the institutional rate. CAPABLE measured the inversion directly: roughly $3,000 in program cost against more than $20,000 in reduced Medicaid spending per participant,[12] with disability reduced across multiple randomized trials.[13] **The money to support older adults at home largely exists already. It is spent later, on worse outcomes, by whoever is left holding the consequence.** Every dollar of that consequence sits on an identifiable balance sheet: a health plan's, a state Medicaid budget's, a health system's, an employer's, or a family's. That is what makes this an addressable market and not only a public health problem.

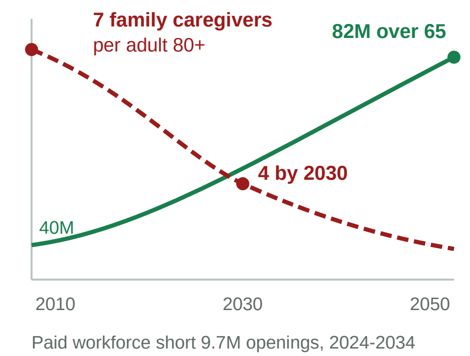

Figure 2. Demand rising while both sources of supply contract.

Why information is not the bottleneck. A family that knows exactly which program it qualifies for still has to complete the application, gather documents, file, wait, respond to requests, and then find a provider with capacity on the date the benefit starts. Existing help stops before that point. Clinicians hand off, information services return lists, and referral marketplaces introduce a provider and are paid on placement. None is accountable for whether care was established, and none can create a caregiver who does not exist.

Two things are therefore required at once: navigation that runs to a confirmed start, and capacity on the other side so the start is possible. **Neither is sufficient alone, and that is the structural claim of this application.** A product that clears only the first two gates delivers family demand to a supply side that cannot absorb it, which is a faster path to the same refusal. We did not choose workforce development as an adjacent opportunity. We hit it as the binding constraint on our own product working.

The product and the path to market. CareNavigator is an eldercare AI navigation system built over an expert-curated national database of aid programs and providers, developed across NIA SBIR Phases I through IIB. It screens a household's needs and means and identifies the aid and services that fit. What this award adds is execution: agents that file the application or send the intake packet, follow up on a schedule, and confirm the date aid or care began. Alongside it, Olera recruits and verifies new caregivers and places them with licensed providers, who hire, train, insure, and supervise them as employees. **Only Caregiver Staffing is sold and priced during this award.** Families use CareNavigator at no cost, and no provider pays to be listed or to receive a family connection. Because no fee is charged for a referral, no steering incentive exists and federally reimbursed providers can participate, which pay-per-referral models cannot accommodate.

Market segments and customers. The beachhead customer is the healthcare and long-term services provider. Roughly 165,000 such organizations operate in the United States,[15,16] and they already spend at least $5 billion a year on marketing and caregiver recruiting, an estimate built entirely from published low-end inputs.[17] They buy from us for an auditable reason: the published median cost of acquiring a caregiver is $520 per hire, and that figure is for word-of-mouth, the cheapest channel an agency has.[18] Table 1 sets our pricing against that benchmark on the provider's own numbers. **We do not ask providers to believe in a new category. We ask them to pay less for something they already buy.** Break-even arrives at 6.3 hires a year, an agency of roughly eight caregivers, so every larger agency is ahead from the first month.

Agency size	Hires per year at 75% turnover	Current cost at the $520 median	Olera at $275/mo, unlimited hiring	Effective cost per hire	Annual saving
15 caregivers	11	$5,850	$3,300	$293	$2,550
30 caregivers	23	$11,700	$3,300	$147	$8,400
50 caregivers	38	$19,500	$3,300	$88	$16,200
100 caregivers	75	$39,000	$3,300	$44	$35,700

Table 1. What a provider pays today to hire a caregiver, and what Olera charges instead.

A second customer class is emerging, and this award produces evidence for it rather than selling to it. Medicare now pays directly for care navigation and caregiver support through the GUIDE model,[19] and Medicare Advantage plans, managed care organizations, health systems, self-insured employers, and long-term care insurers each bear part of the cost of unmet need. The establishment and cost data this award produces is what makes them addressable. **None is tested here and none carries an endpoint in any aim.** The Commercialization Plan models them, with the market sizing and multi-year projections this section does not carry.

Competitive environment and our advantage. Table 2 places CareNavigator against every category of alternative across the five stages of the path to care. The pattern rather than any single row is the argument: **every alternative solves one stretch of the path and stops at a handoff, and the handoffs are where families are lost.** Human navigators are the strongest competitor and cover four of the five stages. They are also scarce, episodic, and cannot be scaled by hiring, which is the same workforce constraint from a different direction. Referral marketplaces charge placement fees, which determines who appears and has drawn sustained criticism of the incentives it creates.[22]

Stage of the path to care	**CareNavigator** *navigation and new workforce*	**Public information** *Eldercare Locator, BenefitsCheckUp*	**General AI assistants** *ChatGPT, Claude, Gemini*	**Human navigators** *social workers, discharge planners*	**Referral marketplaces** *A Place for Mom, Caring.com*
Assesses needs	✓	✓	✓	✓	✓
Identifies aid and insurance	✓	✓	✓	✓	✗
Matches local providers	✓	✗	✗	✓	✓
Carries the case to established care	✓	✗	✗	✓	✗
Adds the caregivers to staff it	✓	✗	✗	✗	✗

Table 2. Coverage of the path to care: the alternatives families use.

Caregiver Staffing competes against a different set of incumbents, and the same pattern holds. Job boards leave every screening decision to the provider, staffing agencies keep the worker as their own employee, and gig platforms fill a shift without filling a role. All three draw from the same constrained pool of people already in direct care. **None of them adds a caregiver to the field,** which is the distinction Key Innovation 1 develops.

Related development efforts in academia and industry. The relevant science is largely settled and what remains open is operational, which is why this is a commercialization award rather than a trial. Unmet daily-activity needs are established as predictors of downstream utilization and placement,[4,5] so we measure care established rather than clinical endpoints. Home-based support trials, principally CAPABLE, show that function-focused support reduces disability with savings well above program cost,[12,13] but none addresses how a family outside a trial finds such support, funds it, and gets it started. CareNavigator is the front door those programs assume exists. Studies of digital caregiver tools, four of them our own, establish that such platforms can be built and measured as usable and accepted;[25,26,27,28] what remains open is whether agents can complete applications and carry a case to a decision. And documented patient-care hours are a requirement for health-professions admission,[23,24] the incentive our workforce pathway uses; whether it converts into durable capacity is what Aim 2 tests.

Hurdles to adoption. The primary hurdle is local density at a cost the market sustains: families, providers, and caregivers must concentrate in the same market, and that market's revenue must cover the cost of bringing them together. Four hurdles follow, each answered by a named aim. Families must trust an AI system with care decisions, so every answer comes from a database audited by a blinded panel of licensed social workers in Aim 1, with acceptance measured under IRB approval in Aim 2. Providers are wary of platforms, so Aim 2 gives them the product free and Aim 3 asks a different set of providers to pay only after the value is demonstrated. New caregivers must be safe and welcome, which is why licensed providers employ, train, insure, and supervise them, and why Aim 2 measures the pathway end to end. And every market must repeat affordably, which self-service tools and a documented playbook address, with acquisition cost measured in Aim 2 and re-measured in each Aim 3 wave.

INNOVATION

Key Innovation 1: capacity created, not moved. Every staffing channel serving this industry, whether an agency, a job board, or a gig platform, competes for workers already inside the direct-care labor market. One provider's hire is another provider's vacancy, and the national shortage does not shrink by a single worker. Olera adds people who were not in the field (Figure 3).

The first population tested is health-professions students, chosen because admission to their programs requires documented patient-care hours. That makes recruitment structurally inexpensive rather than dependent on wage competition. Health professions confer 263,800 undergraduate degrees a year,[23] entry into nursing, medicine, and physician assistant training turns on documented direct patient care,[24] and our own pilot drew more than 900 applicants. New cohorts arrive every semester, so the pipeline refills rather than depleting.

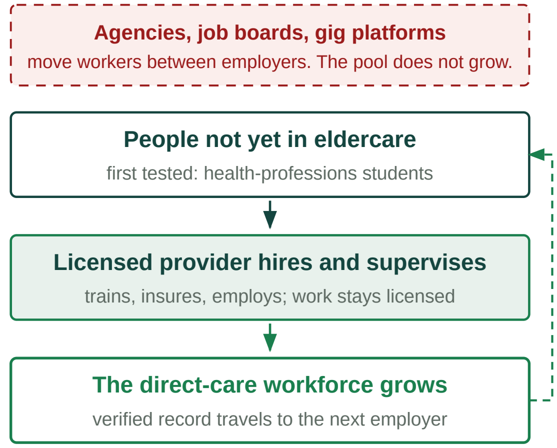

Figure 3. Existing channels move workers between employers. This pathway adds them, and the verified record follows the worker.

The innovation is not the student channel, which any competitor could copy. It is the infrastructure underneath it: the machinery that converts a person who has never worked in eldercare into someone a licensed provider will hire, carrying their verified record of hours, competencies, populations served, supervisor evaluations, and credential status across employers. That record does not exist anywhere today. A caregiver who changes employers starts over, and the new employer re-screens work already verified. That layer is what makes any new worker pool viable, and its value compounds with every shift worked. Nothing in it is specific to students: the pathway is population-agnostic by design, and Aim 2 measures it in one market with a health-professions campus nearby and one without, so the dependency is tested rather than assumed.

Key Innovation 2: execution, not information. The prevailing paradigm is that tools inform and families execute. A family is told which programs it may qualify for and is then left to prepare applications, gather documents, file, wait, respond to requests, and confirm that care began. Those are precisely the steps at which families are lost, and no existing tool is accountable for them.

CareNavigator's agents perform the work and the family decides. **The unit of success changes from information delivered to care established.** That shift makes the outcome measurable, makes it saleable to an organization that bears the cost of unmet need, and makes the system improve: every completed case becomes supervised training data for the next. The multi-agent architecture over a retrieval layer with expert feedback was built and evaluated under Phase IIB. What this award adds is execution and confirmation.

Key Innovation 3: a county-level record of what actually happens. Eldercare information exists in three layers, and almost everyone works from the first one. The open web holds public directories and program rules. It is often neither accurate nor current, and it is equally available to any general-purpose AI system, which is why a chatbot can describe a benefit and cannot tell a family whether their county's waiting list is open.

Beneath that sit local program realities and provider truths that were never published: which office actually processes an application, what a program requires in practice rather than on its form, which providers take which payment sources, and which have capacity. We assembled that layer through field work, expert curation, and direct outreach into more than 72,000 records covering all fifty states, across three NIA awards.

Beneath that sits a third layer that can only be learned by operating: what programs actually decide, county by county, where applications stall, whether capacity existed when it was needed, and what care ultimately started. **No public dataset holds it.** NHATS is a survey. Claims data record what was billed, not what was needed and refused. Neither observes the interval between a family recognizing a need and care beginning, which is the interval in which the system fails them.

Execution is what produces that layer. An agent that files an application and follows it to a decision records the decision; a placement that runs to a first confirmed shift records whether capacity existed. **What this project builds is the first longitudinal, county-level administrative record of what older adults need, what they qualify for, where the system fails them, and whether care was established.** Its commercial value is that a competitor who copies every feature begins with an empty database, and the gap widens with every case we complete. Its scientific value is that the measurement does not exist for researchers or for the state agencies that plan around it, and we commit to publishing from it.

APPROACH

Overall research design. Olera builds two products, CareNavigator and Caregiver Staffing, and during this award only Caregiver Staffing is sold and priced. Everything that could be established without customers has been (Table 7). One question remains, and it is not a research question. **Can a single local market pay for itself, and can we repeat it?**

The award answers that in three stages (Figure 4): engineer the technology, pilot it free in two markets, then pilot it paid in eight new ones.

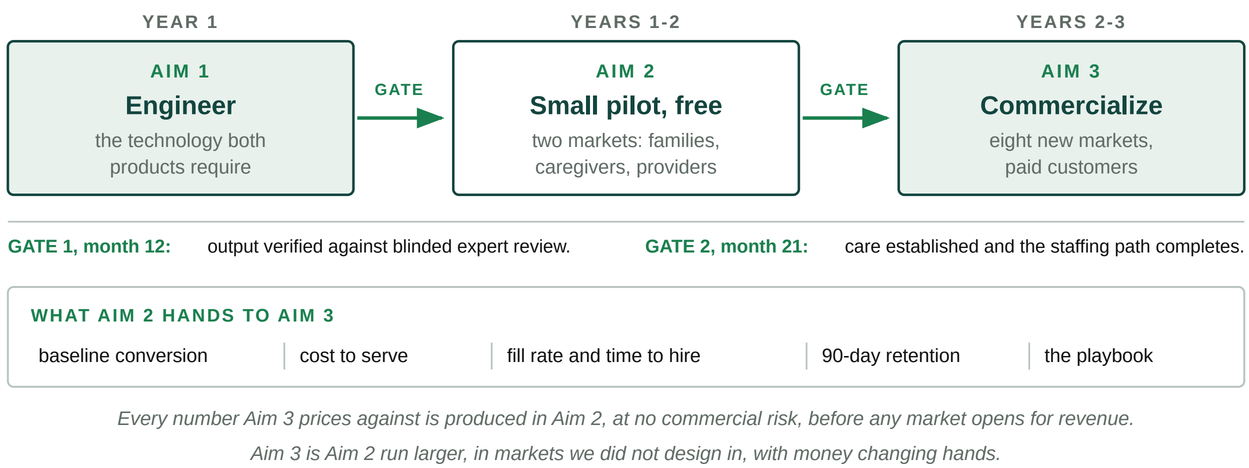

Figure 4. The three aims as one sequence. Each stage carries a gate, and Aim 2 produces every parameter that Aim 3 prices against.

The sequence is the design, not a schedule. **Every parameter Aim 3 prices against is produced by Aim 2 at no commercial risk**, as Figure 4 sets out. Aim 3 is Aim 2 run larger, in markets we did not design in, with money changing hands. Each stage carries a gate, so a failure stops rather than spreads: if Aim 1 misses verification, no market activates; if Aim 2 misses establishment, wave one is held.

Primary endpoints are operational or economic and come from platform, billing, payroll, and employment records. Human subjects research sits inside the aims as Tasks 2.3, 2.5, and 3.4, under Clemson University IRB approval with Dr. Fan as protocol lead. Consent, privacy, data security, and risk assessment for all three studies are detailed in the PHS Human Subjects and Clinical Trials Information form. The project is not a clinical trial: no participant is assigned to receive or withhold navigation, health-related outcomes appear only as secondary observational measures on a cohort that all receives the service, and the Aim 3 price conditions are assigned to markets with purchase as the outcome. Determinations will be obtained rather than assumed, including for the streams we expect to be non-human-subjects research.

Ten markets, and why that number. Table 3 derives the count from what each aim has to measure: two free markets that differ on workforce source, then eight paid markets that let price be tested at four points without confounding price with market type. At roughly $30,000 to enter a market, ten markets is about seven percent of the requested budget. The binding constraint is operating attention, not capital.

What the design requires	Why	Markets
Pilot before charging, in more than one place	One market cannot separate a product that works from a market that happens to work, and the two must differ on workforce source so the campus dependency is tested while it is free to learn	2, free, in Aim 2
Four price points, not two	Two points show which direction demand moves. Four show the shape of the curve, which is what choosing a price requires	4 arms
Two markets per arm, crossed with market type	Price is assigned by market because neighbors compare quotes, so one market per arm means the arm is a market. Within each pair, one campus-rich and one campus-poor	4 × 2 = 8, paid, in Aim 3
Two waves of four	Wave one at month 21 is the only wave with runway to observe 12-month retention. Wave two at month 30 is run as written by staff who did not design it	4 at month 21, 4 at month 30

What the design requires	Why	Markets
Derived design	**Two free pilot markets, then eight paid markets in two waves**	**10 markets, about 7% of the budget**

Table 3. How the number of markets follows from what the aims have to measure.

Specific Aim 1: Engineer the technology and infrastructure both products require. *(Year 1)*

Phase IIB built identification. This aim builds execution, the workforce infrastructure, and the data layer underneath both, and proves the output is good enough to put in front of families before any market opens.

Task 1.1: Execution and follow-up, across aid and services. CareNavigator today screens a household and identifies the aid and services it qualifies for. That much runs nationally. The execution loop is in development and the follow-up loop does not exist. This task builds both. Families are lost at the same two points whether they are securing a benefit or securing care: while assembling what is required, and again while waiting for an answer. For aid, the loops assemble and submit applications, track each program's documentation, and follow the case to a decision. For services they apply the same pattern to a different counterparty: prepare and send the intake packet, schedule the assessment, follow up with more than one provider so a single decline does not end the case, and confirm the date care began. Nothing is transmitted without the family's approval.

Task 1.2: The aid, provider, and outcomes database, and the domain model built on it. The curated record of more than 72,000 aid programs and providers is expanded and re-verified on a rolling schedule, so accuracy does not decay as program rules change. New instrumentation captures what each completed case decided: what was applied for, what was approved or denied, how long it took, and what care started. Model development continues the retrieval-augmented and expert-feedback methods evaluated under Phase IIB, now over a record that includes outcomes rather than rules alone.

Task 1.3: Caregiver workforce infrastructure and the verified experience record. Recruitment and screening workflows, training completion tracking, and the provider-facing placement interface, replacing the manual delivery that capped the pilot. The verified record accumulates hours, competencies, populations served, supervisor evaluations, and credential status, and travels with the worker to any licensed employer.

Task 1.4: Verification against blinded expert review. A panel of licensed clinical social workers, independent of the engineering team and holding no equity in Olera, audits a stratified random sample of cases each quarter. For each audited household the panel determines eligibility blinded to what the platform produced, and for a subset prepares the applications by hand. Both are compared with the platform's output, category by category for eligibility and field by field for applications. Agreement is reported as percent agreement with a 95 percent confidence interval and as Cohen's kappa. Inter-rater reliability among panelists is established before any system output is adjudicated, so panel disagreement is not mistaken for system error. Every material error is reviewed rather than sampled and confirmed by a second reader, and audited households are held out of tuning data. This is internal product verification, not human subjects research, and a determination will be obtained.

Task 1.5: Market selection and preparation. Markets are ranked on existing organic family traffic, provider density, what a household can secure under that state's aid rules, and proximity to a health-professions campus. The two Aim 2 markets are selected from that ranking, with partnerships in place before activation. Year 1 ends with markets ready, not software alone, so the month 12 gate is a launch decision.

<u>Metrics for Success and Quantitative Criteria (Aim 1):</u>

Measure	Criterion	Source of the measurement
Agreement with the blinded expert panel	≥ 85%	Quarterly stratified audit, Task 1.4
Material errors against the expert gold standard	≤ 10%	Same audit
Inter-rater reliability among panelists	$\kappa \geq 0.70$	Established before adjudication
Execution loops complete end to end, aid and services	Both paths	Internal release testing
Prepared workflows carrying a current state and a next step	100%	Platform telemetry
Applications transmitted without family approval	0	Platform audit log

Measure	Criterion	Source of the measurement
Escalations reaching a navigator or named local agency	100%	Escalation log
Scheduled follow-ups sent on their due date, every miss logged	≥ 95%	Platform telemetry
Verified experience record populated for every placement	100%	Workforce system

Table 4. Aim 1 quantitative success criteria.

Gate: the first two criteria must be met at month 12 before Aim 2 activates.

Potential problems and alternative strategies. Program rules change during a three-year award and would degrade accuracy through no fault of the system. The re-verification schedule in Task 1.2 addresses this, and accuracy is reported separately for rules that changed inside the measurement window. If agreement stalls below the gate, we narrow scope rather than widen it: automated execution is restricted to the categories where accuracy is highest, the remainder routed to human-assisted execution, and the boundary reported rather than averaged away. Aim 2 activation is held until the gate is met.

Specific Aim 2: Pilot both products in two markets, at no charge. *(Years 1 to 2)*

With real households, providers, and caregivers concentrated in one place, does care get established, does the staffing pathway complete, and do all three sides accept it? Nothing is charged, so adoption is not confounded with price, and every parameter Aim 3 needs in order to price is measured here first.

Task 2.1: Activate two markets. The two markets selected in Task 1.5 open, one with a health-professions campus nearby and one without. Families arrive through organic search, local partners, and direct outreach. Providers come from the onboarded base and direct outreach and are offered both products at no cost, grandfathered for the life of the award, so no pilot provider faces a purchase decision that could color the answers they give here and Aim 3 tests price only on customers who were never given anything free. Every activation is tagged by source, market, and cohort.

Task 2.2: Measure care establishment. Each household is followed from screening through the identified aid and services to a confirmed start date, with the step at which any household stops recorded. Reported outcomes are the share reaching established aid or care, time to established care, aid dollars secured, drop-off by step, and eligibility accuracy against the accumulating volume of executed cases, which tests whether the record sharpens with use. What counts is fixed before measurement begins: clicks, impressions, and repeat contacts are excluded, an outcome counts once, and every count is reconciled against the outside record of the same event. This is platform telemetry, not human subjects research.

Task 2.3: Validate CareNavigator with family caregivers. *(Human subjects research; Clemson University IRB.)* Twenty-five family caregivers of people living with dementia complete a mixed-methods evaluation of the integrated product, following the User Experience Evaluation in Intelligent Environments framework. Participants are adults caring for a person living with Alzheimer's disease or a related dementia and involved in arranging or paying for that care, recruited from aging and caregiving organizations and our own community, sampled to reflect the range of our users. *Sample size.* Detecting a medium effect on the pre-post usability comparison at 80 percent power and a two-sided alpha of 0.05 requires 15 completers; enrollment is set at 25 against the 35 to 40 percent attrition reported for dementia caregiver studies. In a 60-minute moderated session each participant completes a needs assessment, reviews eligibility matches, prepares one application package, and locates the status of a filed application, thinking aloud while a moderator outside the engineering team records task success against criteria defined in advance, time on task, errors, and assists. Post-use measures are the System Usability Scale and the 12-item trust-in-automation scale, followed by a semi-structured interview on where the system lost them. Surveys at four weeks and three months record whether the participant reached aid or care. Transcripts are analyzed thematically against a shared codebook by two independent coders with inter-coder agreement reported and qualitative reporting against the COREQ checklist. Findings are integrated in a joint display and converted into an engineering backlog that must clear before the month 21 gate.

Task 2.4: Recruit, place, and retain caregivers. Recruitment runs through campus advisors, health-professions organizations, and community channels. Prior direct-care employment is ascertained at intake, before placement, so the share of workers new to the field is measured rather than inferred. Each placement is timestamped from application through screening, interview, hire, and first confirmed shift, which yields time to hire and fill rate. Retention is reported by cohort at 90 days, separately for workers who stay with the placing provider and those who move to another licensed employer carrying their verified record, because a worker who changes employers still represents capacity added to the field.

Task 2.5: Validate the staffing product with providers and workers. *(Human subjects research; Clemson University IRB.)* A sequential mixed-methods implementation study in two phases. *Formative phase.* Approximately 30 provider participants across owner, recruiter, and scheduler roles and approximately 20 placed workers complete role-specific tasks against criteria defined in advance, then a guided interview on workflow fit, supervision and safety, training needs, and priorities for improvement. The target provides at least five users per role, consistent with problem-discovery guidance, and recruitment continues in small batches until no new high-priority issue emerges in two consecutive batches. *Field phase.* The refined product runs through live semester cohorts in both markets with approximately 60 provider accounts and approximately 100 placed workers followed for three months, a target that carries a 20 percent attrition allowance. *Measures and analysis.* The System Usability Scale is the primary usability outcome; the Acceptability, Appropriateness, and Feasibility of Intervention Measures are secondary. Behavioral outcomes are critical-task completion, time to first value, repeat use, and support minutes. Binary outcomes are analyzed with generalized estimating equations using a logit link and an exchangeable working correlation, clustered at the provider organization with small-sample corrections; market is a reporting stratum rather than a model term, and mixed models are pre-specified as a sensitivity analysis. Estimates are reported with 95 percent confidence intervals as implementation estimates, not between-group tests. Interviews follow a guide informed by the Consolidated Framework for Implementation Research and are analyzed by two analysts in a framework matrix with an audit trail. *Safeguards.* Research participation is separated from employment decisions, shift allocation, and any commercial conversation; worker responses are not shared with employers; and withdrawal affects neither platform access nor placement. A stated-preference component, run apart from any sales conversation, seeds the price range Aim 3 tests.

Task 2.6: Measure cost to acquire and cost to serve. Acquisition cost is measured per family and per provider by channel and market. Each channel runs under a budget, an attribution window, a cost ceiling, and a decision rule set in advance; channels that miss their ceiling are closed rather than carried. Cost to serve is measured per established case and per placed worker by time-driven activity-based costing over agent compute, messages sent, navigator time, and support time. Both formulas are fixed before measurement begins and both read from live records. These figures are the denominators of every economic claim in Aim 3.

Metrics for Success and Quantitative Criteria (Aim 2):

Measure	Criterion	Source of the measurement
Households reaching established aid or care	Reported with 95% CI	Confirmed start date, Task 2.2
Outcome ascertainment at the pre-specified, category-specific window	≥ 80%	Task 2.2
Task completion, with drop-off per step, held two cycles running	≥ 90% / ≤ 10%	Task 2.2 funnel instrumentation
Usability, System Usability Scale	≥ 72	Task 2.3, IRB, n = 25
Trust in automation	≥ 5 of 7	Task 2.3, IRB
Placed workers with no prior direct-care employment	Reported	Ascertained at intake, Task 2.4
Worker retention at 90 days	Reported by cohort	Employment and verified-record data
Provider acceptability, appropriateness, feasibility	≥ 4.0 of 5	Task 2.5, IRB
Cost to acquire and cost to serve	Measured, both sides	Task 2.6, time-driven activity-based costing

Table 5. Aim 2 quantitative success criteria.

Gate: care established and the staffing path completing end to end, at month 21, before wave one opens.

Potential problems and alternative strategies. Worker retention is the most consequential risk in this application. If people new to the field leave within 90 days, providers will not pay a second time and Aim 3 weakens no matter how well navigation performs. Two responses are set in advance. Retention is reported both with the placing provider and in direct care with any licensed employer, because those answer different questions and only the second speaks to capacity added. If retention with the placing provider is poor while workforce retention holds, the product emphasis shifts from placement to the verified record itself, which providers value as reduced screening burden. Campus seasonality could open hiring gaps between terms, so cohorts are staggered across the two markets. If the market without a campus produces no caregivers, that is a finding rather than a loss: the family side continues there and Aim 3's market selection is revised toward the conditions that worked, with the boundary stated. If family volume is too low for providers to experience value, family acquisition is reallocated before any provider-side conclusion is drawn.

Specific Aim 3: Pilot the paid products in eight new markets. *(Years 2 to 3)*

Aims 1 and 2 establish that the products work. This aim asks whether providers pay, whether the unit economics hold after real costs, and whether the playbook repeats in markets we did not design in. That is the condition on which the business, and any private investment in it, depends.

Task 3.1: Open eight new markets in two waves of four. Wave one opens four markets at month 21, wave two four more at month 30. Both run the playbook documented in Aim 2, and wave two is executed by staff who did not write it, with every deviation logged. Waves are compared on time to first provider, cost per activation, and the self-service share of activations, which tests whether the model repeats affordably rather than merely repeats. These providers are new customers who were never given the product free, so conversion carries no prior-gift confound.

Task 3.2: Set price and packaging under real billing. Four price points are tested, two markets each. Two points would show which direction demand moves; four show the shape of the curve, which is what choosing a price requires. Within each pair, one market has a health-professions campus nearby and one does not, so price is crossed with workforce source rather than confounded with it. Candidate prices are seeded by a Van Westendorp price-sensitivity survey of approximately 120 provider decision-makers screened for purchase authority and drawn from both market types, whose cumulative response curves define the acceptable range and the points entered into the arms. That survey runs under IRB approval, apart from any sales or renewal conversation. The arms are anchored on prices providers have already paid us, approximately $275 per month and approximately $150 per placement. Conditions are assigned by market because providers in neighboring markets compare quotes; markets are paired non-adjacently and a contamination monitor runs in the offer workflow. Price points, package variants, the primary contrast, the follow-up horizon, and the decision rule are pre-registered before any outcome is reviewed. The primary outcome is paid conversion within 60 days of offer, analyzed with generalized estimating equations at the account level with market as the cluster and small-sample corrections. A biostatistician sizes the arms against the conversion rate measured in Aim 2 rather than an assumed one, and the enrollment window is set so key proportions carry 95 percent confidence half-widths of no more than seven points. With eight markets, arm-level contrasts are estimated with confidence intervals rather than significance-tested, which is the honest use of this design. A pre-registered interim analysis at month 30 drops a dominated arm and reallocates to wave two. The decision rule selects the price maximizing expected 12-month revenue per account rather than conversion alone, because a price that converts and then churns is worse than one that does neither.

Task 3.3: Measure unit economics, retention, and cross-side value. Every figure comes from live billing, payroll, and cost records. Acquisition cost includes spending on non-converters. Monthly net account margin is revenue less the cost to serve defined in Task 2.6. Retention is estimated at 3, 6, 9, and 12 months, and time to churn with discrete-time survival models on account-month records matching the billing cycle, treating voluntary cancellation, uncured payment failure, and downgrade as competing events, with cause-specific hazards and cumulative incidence reported. Restricted mean survival time at 12 months gives expected account lifetime, and lifetime times monthly net margin gives lifetime value with a confidence interval, so the payback and lifetime-to-acquisition-cost criteria carry measured uncertainty rather than point estimates. *The two-sided test.* That family demand makes a provider account more valuable is a claim this application makes, so it is tested. Provider value, retention, and margin are modeled against a prior-period measure of consented, qualified family demand in the same market rather than same-month demand, with the adjustment set capped at six terms drawn from provider size and type, baseline activity, acquisition channel, and seasonality, market as a random effect, and the provider organization as the clustering unit. Per-market contribution margin is reported after the cost of serving that market's families, the number that decides whether a market pays for itself.

Task 3.4: The Aim 2 instruments at scale. *(Human subjects research; Clemson University IRB.)* The implementation measures from Task 2.5 are administered across the paid cohort, with a baseline at enrollment and follow-up at four weeks and three months or after a key event such as activation, downgrade, or churn. Baseline perceived value enters the Task 3.3 survival models as a churn predictor specified in advance. After the interim review, approximately 30 providers are interviewed, sampled purposively across four groups: never converted, converted then churned, retained with low use, and retained with high use. Churn subtype, voluntary or payment failure, is recorded and used in sampling. We target at least six providers per group and document the stopping rule. Interviews examine what was purchased, what value was perceived, how price was judged, and why accounts lapsed, and run apart from sales, support, and renewal contact. Providers who did not convert are offered a shortened format, because a sample of satisfied customers would answer the wrong question. The person-level participant and the account-level economic unit are defined separately, and the joint display converts into the packaging decisions available to Task 3.2.

Task 3.5: Independent rebuild and the investor evidence package. An independent analyst under subcontract, working from the pre-specified analysis plan and the raw billing, payroll, and cost records, rebuilds revenue, acquisition cost, cost to serve, margin, retention, and per-market profitability without access to our operating model. The analyst does not set price or packaging. Discrepancies between the rebuild and the operating model are investigated and reported rather than silently reconciled. The package assembles the confirmed model into two forward trajectories, expansion funded by reinvested margin and accelerated expansion under private capital. It is the evidence base for the Commercialization Plan's fundraising strategy, which carries the market sizing, price list, and multi-year projections in full.

Metrics for Success and Quantitative Criteria (Aim 3):

Measure	Criterion	Source of the measurement
Paid conversion within 60 days, by price arm	Estimated with 95% CI	Task 3.2, four arms across eight markets
Lifetime value against acquisition cost, 12 months	≥ 3 : 1	Task 3.3, restricted mean survival time
Payback period on acquisition cost	< 12 months	Task 3.3
Per-market contribution margin after cost of serving families	Positive	Task 3.3
Retention at 3, 6, 9, and 12 months	Reported by wave	Discrete-time survival, competing events
Cost per activation, wave two against wave one	Falling	Task 3.1
Provider value against prior-period family demand	Estimated with 95% CI	Task 3.3, market random effect
Independent rebuild delivered, discrepancies reported	Delivered	Task 3.5

Table 6. Aim 3 quantitative success criteria.

Potential problems, alternative strategies, and the stop rule. Contamination across neighboring markets is the standard objection to market-level assignment; the pre-registered mitigations are non-adjacent pairing, a contamination monitor in the offer workflow, and account-level assignment specified in advance as a sensitivity analysis. If conversion is too low at every price for the arms to be distinguishable, the interim analysis reallocates and the result is reported as a finding. If price sits below cost to serve, the pre-registered alternatives run in order: improve the product where the Task 3.4 interviews locate the gap, reduce acquisition and serving cost, repackage, and re-test in wave two. If margins are positive but below what a market needs to fund itself, no further markets open until the cost structure improves. If wave two slips, wave one still carries the pricing, unit economics, and 12-month retention results.

The stop rule. If, at the month 30 interim analysis, fewer than 40 percent of placed workers remain in direct care at 90 days **and** paid conversion is below 20 percent at every offered price, we will report the provider-funded model as disconfirmed, hold wave two, and redirect the remaining effort to completing and publishing the analysis. A commercialization program should fund projects willing to find out they are wrong.

If staffing revenue alone proves insufficient. One product may not carry a market. We expect it will: two-thirds of home-care providers turned away business in 2023 for lack of staff, and our price sits far below what they already pay to hire. If the measured economics fall short, this award still produces assets that support other revenue without new invention: the provider growth tools already operating today, an employer-paid caregiver benefit on the same navigation product, contracted navigation for risk-bearing organizations on the GUIDE precedent, and aggregate market intelligence from the outcomes record. **None is tested here and none carries an endpoint.** The Commercialization Plan models them.

Technical assistance and project oversight. Three external providers carry defined deliverables. **Clemson University**, through Dr. Fan as Co-Investigator at 25 percent effort, designs and executes the human subjects studies in Tasks 2.3, 2.5, and 3.4 and holds the protocols, integrated through quarterly protocol review and the data-management procedures specified in the subaward. **An independent financial analyst** performs Task 3.5, and **pricing and biostatistical consultation** supports the pre-registration and arm sizing in Task 3.2. The Principal Investigator holds integration authority and every gate decision, against a quarterly milestone review of Tables 4 through 6.

Regulatory plan. No federal premarket pathway applies. CareNavigator is navigational and administrative software that does not diagnose, treat, or make clinical decisions, so it is not a medical device and nothing here waits on a federal approval. Four regimes govern operations: family data under HIPAA-aligned safeguards; referral practice, where charging no fee means no steering incentive exists; employment, training, insurance, and supervision of every placed worker, which rest with the licensed provider rather than with Olera; and any joining of an external record to ours, which requires a data-use agreement and an IRB determination.

Timetable and what the award leaves behind. Figure 5 gives the quarter-by-quarter schedule and the four decision points, each of which holds the next stage until its gate is met. At award end the independent rebuild is delivered.

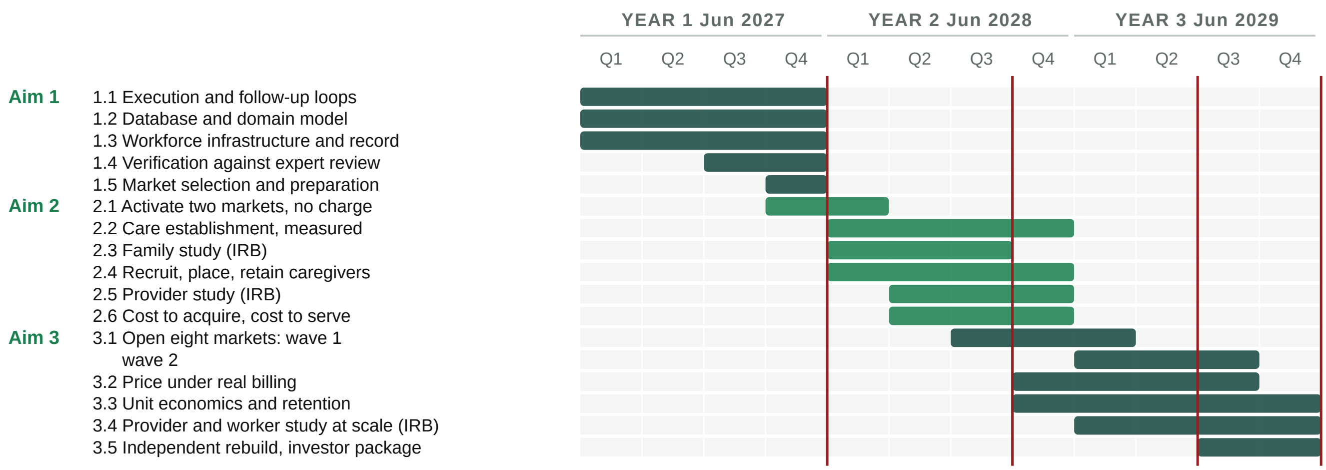

Figure 5. Three-year timetable, showing the sequencing among the aims, staged market entry, and the four formal decision points.

Ten markets and one priced product is a deliberately small footprint for an award this size, because the markets are an instrument and not the deliverable. We are not buying revenue with this grant. We are buying the four things that let someone else fund the revenue: **a product verified against blinded expert review, a price chosen by experiment rather than assumption, unit economics an outside analyst rebuilt from our records, and a market-entry playbook run as written, in markets we did not design in, by people who did not write it.** Those transfer to the next hundred markets. A larger footprint bought with the same dollars would produce more revenue and less evidence, and evidence is what we do not have.

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Development status of the technology. CareNavigator operates in production today. Families screen their needs and means, receive matched aid programs and providers drawn from an expert-curated national database of more than 72,000 records covering all fifty states, and receive AI-drafted, expert-approved guidance. The multi-agent navigation layer, developed and evaluated under Phase IIB using retrieval-augmented generation, parameter-efficient fine-tuning, and reinforcement learning from expert feedback, is in production integration during the remaining Phase IIB year. Execution and follow-up, the capability Aim 1 builds, do not exist in the product today and are not funded by any existing award.

What prior funding established. The technology was developed across NIA SBIR Phase I/II Fast-Track and Phase IIB awards (1R44AG074116), scored at 20 and 25. The Phase IIB review assessed the platform's commercial potential as extremely high, citing its ability to help caregivers access eligible funding. Table 7 records what each activity established and which funding produced it. Two of the five entries were funded outside the Phase II scope, using I-Corps support and the company's own capital.

Risk	Retired by	Evidence
Technical Could the data be assembled?	NIA Phase I to IIB	An expert-curated national database of more than 72,000 aid programs and providers across all fifty states, and a multi-agent eldercare AI now entering production integration.
User Would families accept it?	NIA Phase I to IIB	Four peer-reviewed evaluations with family caregivers: usability 4.57 of 5; acceptance 5.83 of 7 after four weeks of use (n = 65); multi-agent version 5.73 of 7 (n = 31, in preparation).
Demand Would families come, and at what cost?	Beyond Phase II scope; company funds	Organic traffic grew from roughly 50 visits a day in 2023 to more than 500 today, about 15,500 a month, from nearly every county in the country at near-zero acquisition cost.
Supply Would providers participate?	I-Corps and company funds	More than 300 customer-discovery conversations with owners and operators through NIH and NSF I-Corps. More than 700 providers have since claimed a listing at no charge, roughly 150 more each month.
Workforce and willingness to pay Could we recruit caregivers, and would anyone pay?	Beyond Phase II scope; company funds	A student caregiver pilot drew more than 900 applicants and placed more than 20 into provider jobs. Four providers trialed the staffing product and three paid, at roughly \$50 to \$275 per placement and \$275 per month, across multiple semesters. Delivery was manual, which capped scale; Aim 1 removes the cap.
The risk that remains Can one local market pay for itself, and can we repeat it?	Not yet retired	Measurable only with real customers paying real prices over long enough to observe churn. This is the uncertainty the three aims remove.

Table 7. Commercialization risks retired to date, the funding that retired each, and the risk that remains.

What the operating record already shows. Among providers with a claimed account, platform activity runs about fifteen times higher on the day a family lead arrives than on days without one, and a fifth of all growth-tool views fall in the twenty-four hours after a lead. **That is the cross-side effect Task 3.3 tests formally, already visible in behavior.** The provider tools launched nationwide in July 2026 with no paywall, so there is no pricing history to read and Aim 3 sets the first one. The staffing model is also replicating with little staff effort: a committed provider and a university advisor are prepared for a fall pilot in Indiana, seeded by one coordinator.

Ongoing Phase IIB work this award does not re-fund. The remaining Phase IIB year completes production integration of the agent system and continues database expansion. This application begins where that work ends. No budget line here supports agent development already funded under Phase IIB; Aim 1 builds execution, follow-up, confirmation, and the workforce infrastructure, none of which is in scope for the existing award.

The team and the endpoints it works to. Dr. Falohun was Principal Investigator on both prior NIA awards and delivered against their milestones. Dr. DuBose, a physician with an MBA, leads product and market operations and built the organic acquisition channel and the provider pipeline described above. Dr. Fan, at Clemson University, is Co-Investigator at 25 percent effort and leads all human subjects protocols; Dr. Marcia Ory of Texas A&M University advises on study design. Both are co-authors on the evaluation record cited here. Olera has also been evaluated by Ziegler, the leading underwriter of financing for nonprofit senior living providers, and by Equitage Ventures, an early-stage fund dedicated to the aging economy. **That diligence set the commercial endpoints in Tables 4 through 6, so the milestones here are the ones our prospective investors said they would need.**

The question that remains. Nine years of federal and company investment answered everything that could be answered without customers. Whether a local market pays for itself, and whether the result repeats, has to be measured with real money changing hands, over long enough to observe churn. **That is what the three aims do.**

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